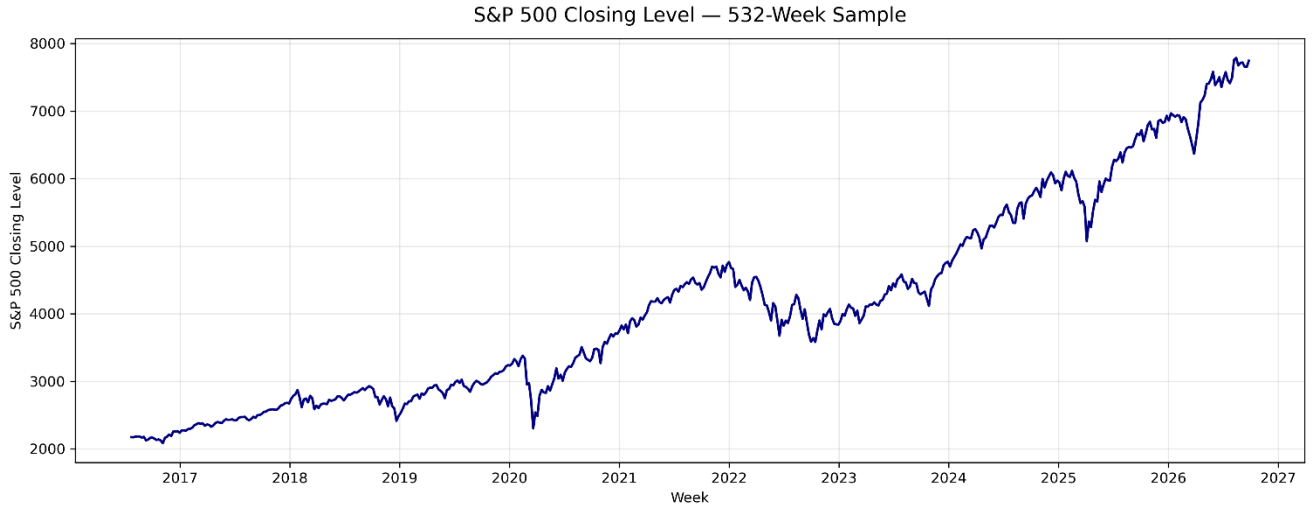
Final Report: S&P 500 Market Performance Model

Model Rationale & Problem Statement – The purpose of the model is to provide guidance on the future performance of the widely followed Standard & Poor's 500 Stock Index. Given a rational means of forecasting the prospective performance of the Index, how could either additions or omissions in investment increase returns, mitigate risk, or assume equivalent returns with less at-risk capital? While outperforming the index, or generating alpha, is difficult, a separate, but meaningful measure of effective portfolio management is achieving similar returns, with less total market exposure. This serves to maximize use of investable capital and minimize losses in market downturns as less capital is at-risk.

The specific application of the model involved the investment needs of a family office. The family office struggled to achieve consistent measures of return from publicly traded equities (stocks) and the timing of their investment strategies. Though the investment managers have deep market experience, they lacked a predictive model to gauge the future returns of the S&P 500 (dependent variable), and considered macro economic, timing, and valuation indicators (independent variables) as key to their future modeling and investment success. The criteria/measure of success is meeting or exceeding the current market returns of the S&P 500 (in this case, reducing the delta/spread between current portfolio returns and those of the broader market), or meeting the Index returns with reduction of risk. With the advent of the model, an indication of near-term stock market returns will allow the family office to add or sustain current levels of market risk. Thus, giving them predictive market insight and the opportunity to change course in their near-term investment strategy.

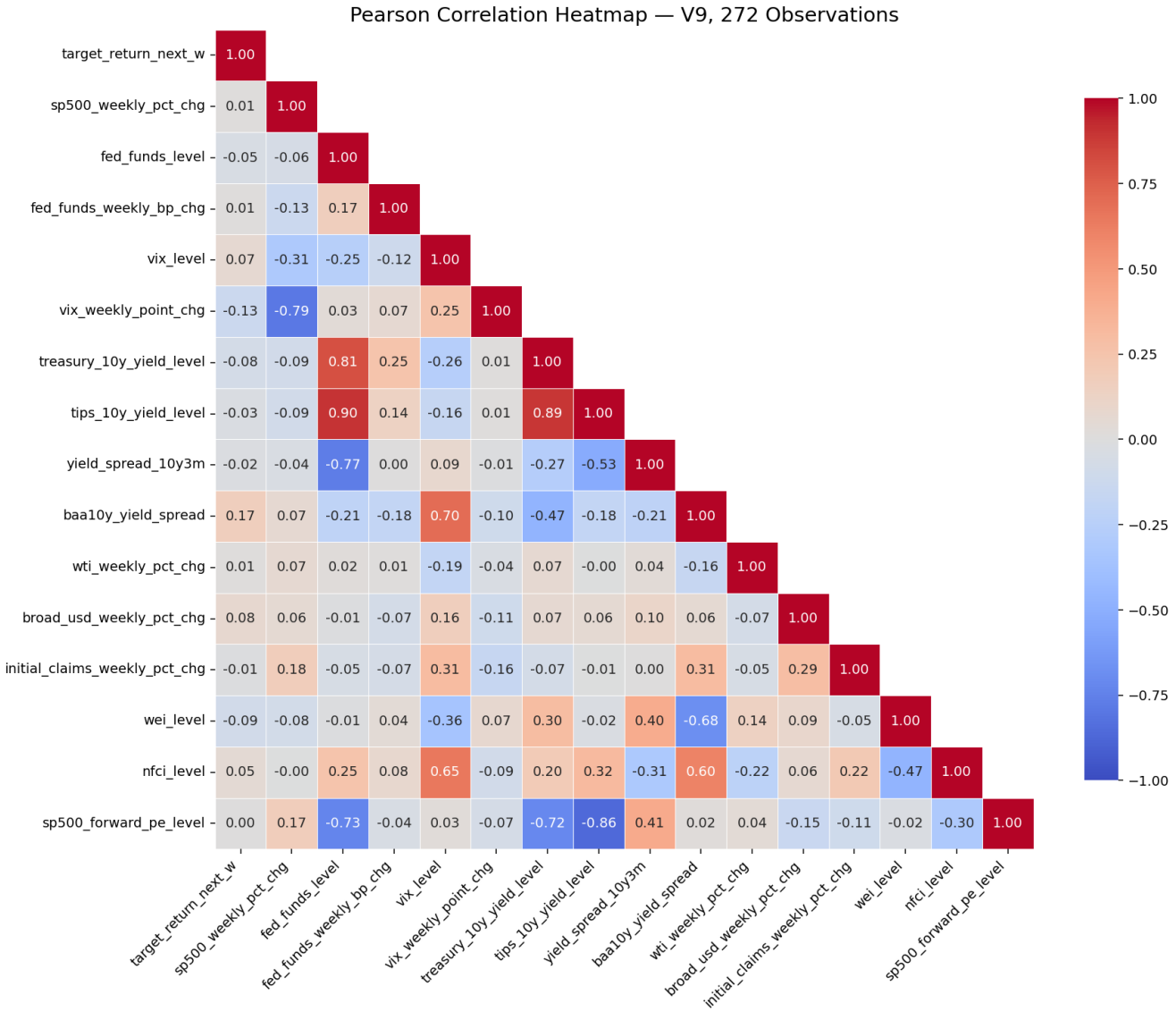


Data Wrangling – The study used publicly available data segmented into three groups: macro-economic, timing, and valuation indicators. The independent variables from these series totaled fifteen, with one also used as the dependent variable (the actual performance of the Index over the period of study). Data was either downloaded directly from an open-API (application programing interface), or compiled through "screen scrapes" (data collected from web pages, but not otherwise saved as a list/table). The underlying data can be characterized as high-frequency as it is released on a weekly basis by the Federal Reserve Bank of St. Louis (data released per "FRED," or Federal Reserve Economic Data), or, in the case of one series, available from FactSet, a capital markets data source specializing in measures of equity valuation. The fifteen data sets that comprise the model's independent variables have 100's or 1,000's of weekly data points spanning back to the 1980's. As FactSet data was missing for the earlier periods, we limited the inputs to the following:

Purpose	**Observation Dates**	**Weekly Count**
Training	January 5, 2018 – March 17, 2023	272
Testing	March 24, 2023 – September 25, 2026	184
Total Weeks Trained and Tested		**456**

Exploratory Data Analysis (EDA) – Exploratory Data Analysis examined predictor distributions, patterns over time, and relationships with next-week S&P 500 returns. Scatterplots and a Pearson correlation heatmap based on 272 training observations showed weak individual linear relationships with the target, although some predictors were strongly correlated with one another. These findings informed further investigation of lagged predictors and threshold features. Weak Pearson correlations did not rule out nonlinear or delayed relationships.

In-Depth Analysis – The training process included plotting the fifteen independent variables (original predictor features) against the S&P 500 Index next-week return (see Appendix A: Individual Predictor-to-Target Relationship for the 272-Week Training Period). The effort was to evaluate the individual independent variables against the dependent variable as per fitted linear regression, or an effort to determine which of the variables exhibited heightened positive or negative linear relationship. Further, a Pearson correlation study quantified the strength and direction of each linear relationship (see below heat map). The results were muted; few of the independent-to-dependent variable pairings yielded signs of linear correlation, or, that, the independent variable provided predictive value for next week's Index movement. The results of the Pearson correlation study over the 272-week training period is presented below. While many of the relationships can be characterized as flaccid, including, for example, forward earnings (sp500_forward_pe_level) to market performance for the next week (target_return_next_w), practical market observation suggests such disadvantaged 0.0 correlation is fallacious. Given the scope and time of this project we add this to our future research list.



While certain of the original variables were moderately correlated with one another, their overall correlation with next-week S&P 500 returns were weak, or they didn't offer positive predictive value of next week's returns. This prompted further investigation of lagged predictors and threshold features.

A lagged time series (lagged predictors) was added to our input slate to determine if a lower mean absolute error (MAE) was possible. It was –the following results were recorded from lagged sets of independent variables—with the two-week period reducing the MAE and retained.

Model Test Results* Adding X-Week Old Values as Additional Inputs				
Weeks Used in Lag Test	**Features Evaluated**	**MAE Before Lag Testing**	**MAE With Lag Testing**	**Result**
One week	15 original predictors +16** one-week lagged predictors	0.0222	0.0232	Error increased; excluded
Two weeks	15 original predictors +16** two-week lagged predictors	0.0222	0.0214	Error decreased; retained
Four weeks	15 original predictors +16** retained two-week lags +16** four-week lags	0.0214	0.0216	Error increased; excluded
*Results using five chronological cross-validation folds within the 272-observation training sample. **The gross price of crude oil per West Texas Intermediate (WTI) was added as a sixteenth variable.				

In addition to data segmentation, which resulted only as a conceptual framework for organization (it ultimately did not serve to help optimize results), feature engineering was employed to enhance model behavior/predictive quality. These thirty-two threshold features (see Appendix A2: Model Feature Set) represent market conditions where there is elevated risk or reward, trend or stagnation, and valuation peaks or troughs. Not all of the original 15 independent variables had threshold features added. As a whole, the features applied provided the following benefit to the predictive quality of the model. The table below illustrates the MAE before and after the threshold features were added to the model.

Model Test Results* Adding Threshold Features			
Threshold Features Evaluated	**MAE Before Addition**	**MAE After Addition**	**MAE Reduction**
All 32 threshold features added to the original predictors and retained two-week lags.	0.0214	0.0213	~0.0001

For the purposes of model training, we validated four models as noted below, with Gradient Boosting providing the best after model tuning results. That is to say that Gradient Boosting was selected because it had the lowest validation MAE among the tuned models, without an excessive training-versus-validation gap. However, none of the tuning models ultimately delivered better results than the baseline MAE for the test group. Given the directive to adopt a tuning model, Gradient Boosting was retained and used for the testing period.

Tuned Model Results Using Four Methodologies		
Model	**Baseline MAE Per Historical Mean**	**MAE After Model Tuning**
Gradient Boosting	0.0201	0.0202
Elastic Net	0.0201	0.0203
Ridge	0.0201	0.0204
Random Forest	0.0201	0.0208

Investment Simulation – Despite modest results in proving causal relationships between the independent and dependent variables, and the model's predictive ability to provide a specific return for next week's S&P 500, a silver lining was realized in the model's overall signal to either invest or not invest. The simulation went as follows: based on the model's positive or negative outlook for next week's S&P 500 change, an investment of $100, or no investment, was tested. Absent the first week where a "forced" investment was made (the transition/bridge period from training to testing), the 184-week period was adopted for simulation of either invest, or don't invest. During the period, a total of $13,200 was invested per the accumulation of positive weekly signals. The money-weighted annualized return of these accumulated investments totaled 18.86% versus the broader market's 18.84% for the same period. While this does not constitute above market returns, or alpha, it does translate into equivalent returns with fewer total dollars contributed, thus, a more efficient investment strategy with the potential for less total losses during market drawdowns as we have recently witnessed.

Consideration	Invest: Predict Positive Returns for S&P 500*	No Investment: Predict Negative Returns for S&P 500*
Investment Periods	132**	52
Cumulative Investment	$13,200	$0
Returns	18.86%***	N/A
*Period of March 24, 2023 – September 25, 2026 / Final valuation date: October 2, 2026 **131 positive-signal contributions + 1 forced contribution for the first period. ***Money-weighted annualized return		

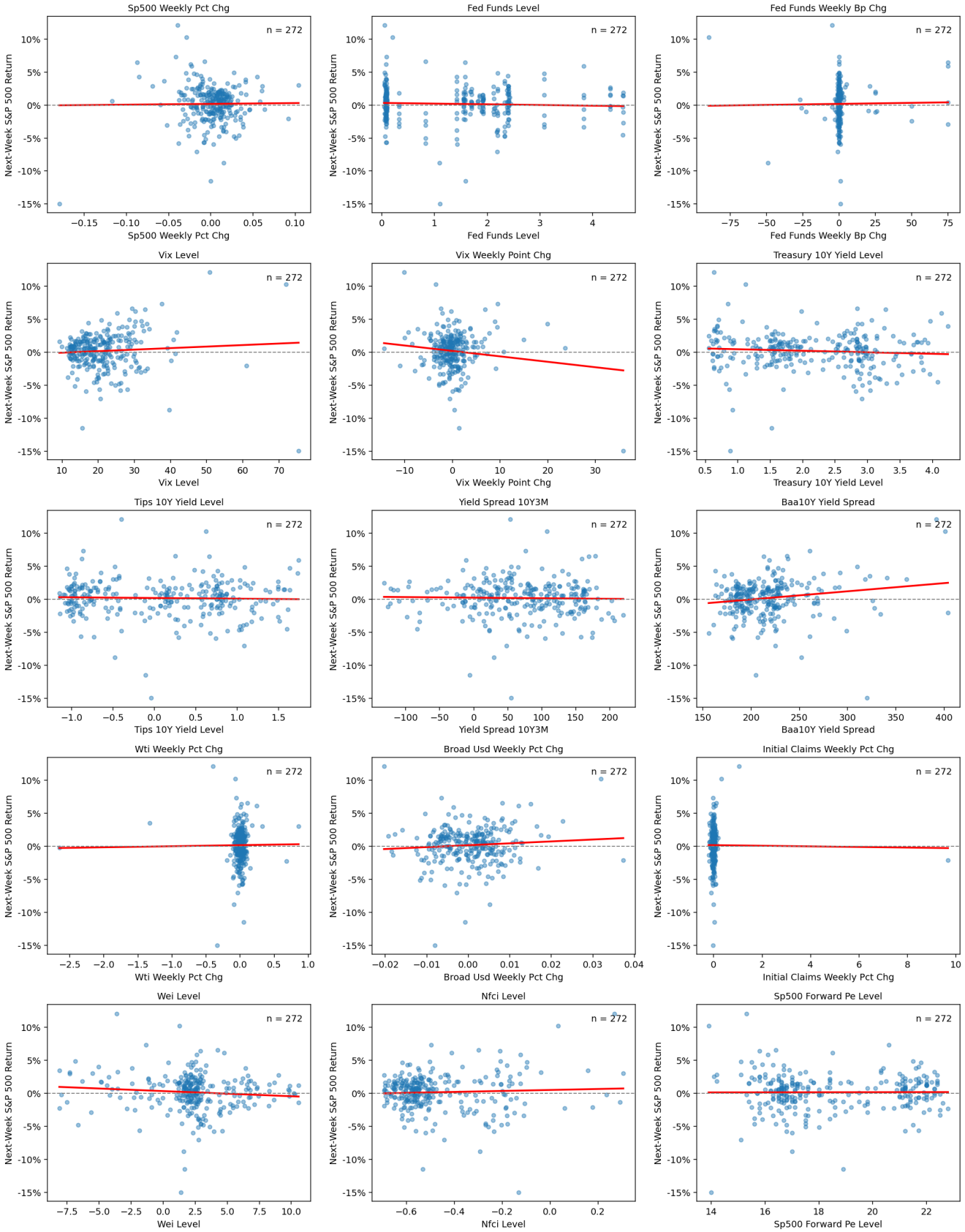
Takeaways – An assessment of findings from the modeled returns of next week's S&P 500 returns includes the following:

- The influence of independent variables to the dependent variable are weak; our model selected 15 independent variables for assessment of predictive or linear relationship to the dependent variable, with few providing actionable value. Or, the movement of our 15 independent variables either individually, or collectively, had little bearing on next week's S&P 500 index returns.
- Lagging our indicators by two-weeks provided benefit in the form of a lower modeled MAE, though our selected threshold features only elevated model performance nominally. Tuned model results using four methodologies was actually negative, and, therefore, the best available, Gradient Boosting, was retained to meet the overall model assignment.
- The model did provide value as a means of signaling either "invest" or "don't invest" for the next week. An investment simulation using the signal proved adequate in matching overall returns for the testing period, but achieved such return with 28.2% less capital contributed. The benefit of less deployed capital would be realized in market downturns.

Future Research – An updated/second version of the model would incorporate the following changes, additions, and return analysis:

- Greater diligence in the selection of first derivative market signals. Examples include using spot market fuel costs rather than the cost of their feedstock, crude oil.
- Use of more direct measures of financial/capital market stress like the Chicago Federal Reserve Financial Conditions Index (NFCI).
- Greater focus on investment decisions and less on gains (losses) for the S&P 500 Index as a whole. This change would allow for withdraws of investment in the index, not just "invest" or "don't invest."
- Segmentation of independent variables, and their weightings thereof, for use in the model. This iteration did not sufficiently test such segmentation and weightings.
- Use of Index moving averages to provide a tighter range of investment criteria.
- Further research into the predictive relationship between forward earnings assessments and future returns of the S&P 500 (our initial research suggests 0.00 correlation while intuition suggests a higher correlation between the two).

Appendix A: Individual Predictor-to-Target Relationship for the 272-Week Training Period

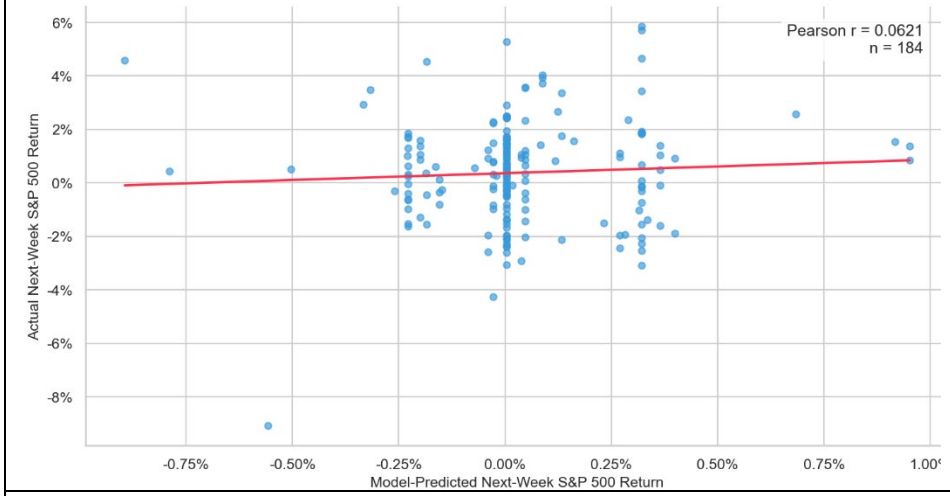
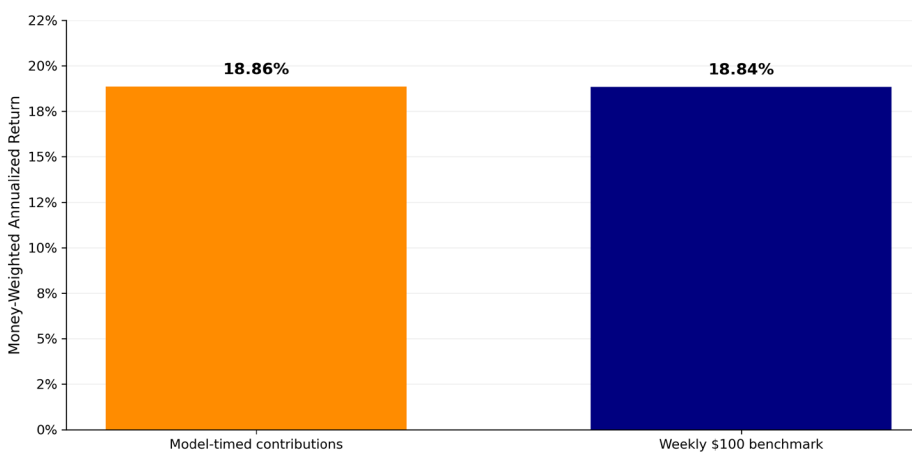


Appendix A2: Model Feature Set

Model Engineered Features: Original, Lagged & Threshold			
Variables	**Original Predictor Features**	**Lagged Predictor Features**	**Threshold Features**
Series1 - S&P 500 Next-Week Return*	No (dependent variable)	N/A	N/A
Series2 - S&P 500 Stock Index**	Yes	By 2-Weeks	Returns -2%/+2%[2]
Series3 - Federal Funds Interest Rate	Yes	By 2-Weeks	None Retained
Series4 - Federal Funds Interest Rate Chg	Yes	By 2-Weeks	None Retained
Series5 - Volatility Rate	Yes	By 2-Weeks	Above 25.0[1]
Series6 - Volatility Rate Chg	Yes	By 2-Weeks	Change in points: -5 /+5[2]
Series7 - US Treasury 10 Year Note Yield	Yes	By 2-Weeks	None Retained
Series8 - US Treasury 10 Year Note Yield***	Yes	By 2-Weeks	None Retained
Series9 - US Treasury Yield Difference	Yes	By 2-Weeks	< 0.0 basis point differential[1]
Series10 - Corporate Bond Yield – 10Y Note	Yes	By 2-Weeks	>250 basis point differential[1]
Series11 - Crude Oil Price	Yes	By 2-Weeks	Change in percent: -5%/+5%[2]
Series11B – WTI Price Level	No	By 2-Weeks	None Retained
Series12 - US Dollar Index	Yes	By 2-Weeks	Change in percent: -1%/+1%[2]
Series13 - Weekly Jobless Claims	Yes	By 2-Weeks	Change in percent: -5%/+5%[2]
Series14 - Weekly Economic Indicator Index	Yes	By 2-Weeks	< 0.0 index value[1]
Series15 - Financial Stress Indicator	Yes	By 2-Weeks	None Retained
Series16 - S&P 500 Weekly Forward 12-Month P/E	Yes	By 2-Weeks	Below 16.0 / Above 20.0[2]
Totals	**15**	**16**	**32**

Notes:
*As a dependent variable
**As an independent variable. All other rows are independent variables.
***10-year TIPS yield, or real yield implied by 10-year Treasury Inflation-Protected Securities.
[1]Flag and excess
[2]Flag and excess for each

Appendix A3: Model Prediction Accuracy (Pearson Correlation Method) and Model Investment Simulation

Prediction Scatterplot Gradient Boosting: Predicted Vs. Actual Next-Week Returns	Investment Simulation Money-Weighted Annualized Return Comparison
	
Timeframe: Testing period of 03/24/23 – 09/25/26 Data: V9 Data Source Results: Pearson r = 0.0621	Timeframe: Testing period of 03/24/23 – 09/25/26 Data: V9 Data Source Results: 132 model contributions Vs. 184 weekly benchmark contributions

Final Model Metrics: S&P 500 Market Performance Model

Model & Data

- Final model: Gradient Boosting regression (GradientBoostingRegressor).
- Authoritative modeling notebook: SP500_Capstone_Modeling V3.ipynb.
- Companion EDA notebook: SP500_Capstone_EDA V2.ipynb.
- Dataset: Springboard - Capstone Two Data Series V9 (10 07 26).
- Source workbook: Springboard - Capstone Two Data Series V9
- (update for missing forward PE values) (10 11 26).xlsx.
- The nine approved forward-filled S16 forward-P/E additions are included.
- Target: target_return_next_w, the S&P 500 return from the current Friday
- close to the following Friday close. The target is not an input feature.

Evaluation Sample

- Training observation dates: January 5, 2018-March 17, 2023; n = 272.
- Test prediction dates: March 24, 2023-September 25, 2026; n = 184.
- Total model input observations: 456.
- The final test target ends October 2, 2026.
- October 2, 2026 is also a separate forecast-only input observation;
- that forecast observation is excluded from the 456 and all test metrics.
- Validation: five chronological folds within the training sample.
- Primary model-selection metric: mean absolute error (MAE); lower is better.
- The evaluated model is the same fitted model used for the investment simulation; it was fitted on the 272 training observations.

Final Model Configuration & Hyperparameters

- learning_rate = 0.03
- max_depth = 1
- min_samples_leaf = 5
- n_estimators = 100
- random_state = 42
- Preprocessing: median imputation fitted within each training fold.
- The Gradient Boosting pipeline does not include feature standardization.
- Learned parameters: fitted tree split rules and leaf values estimated
- from training data. No predictor or threshold weights were set manually.
- The notebook contains the fitting procedure; this file summarizes model
- configuration and results rather than serializing the fitted trees.

Final Features: 63 Total –15 original predictors + 16 two-week lags + 32 threshold features.

A. **Original Predictors (15)**

01. sp500_weekly_pct_chg - S&P 500 weekly percentage change
02. fed_funds_level - Federal-funds rate level
03. fed_funds_weekly_bp_chg - Federal-funds weekly basis-point change
04. vix_level - VIX level
05. vix_weekly_point_chg - VIX weekly point change
06. treasury_10y_yield_level - 10-year Treasury yield
07. tips_10y_yield_level - 10-year TIPS yield
08. yield_spread_10y3m - 10-year/3-month Treasury yield spread
09. baa10y_yield_spread - Baa/10-year Treasury yield spread
10. wti_weekly_pct_chg - WTI weekly percentage change
11. broad_usd_weekly_pct_chg - Broad U.S. dollar weekly percentage change
12. initial_claims_weekly_pct_chg - Initial-claims weekly percentage change

Final Features: Original Predictors (continued)
13. wei_level - Weekly Economic Index level
14. nfci_level - National Financial Conditions Index level
15. sp500_forward_pe_level - S&P 500 forward 12-month P/E level

B. **Two-Week-Lagged Predictors (16)** – Each is an additional input from two calendar weeks before the prediction date; the original current-week predictors remain in the feature set.
01. S&P 500 weekly percentage change, lagged two weeks
02. Federal-funds rate level, lagged two weeks
03. Federal-funds weekly basis-point change, lagged two weeks
04. VIX level, lagged two weeks
05. VIX weekly point change, lagged two weeks
06. 10-year Treasury yield, lagged two weeks
07. 10-year TIPS yield, lagged two weeks
08. 10-year/3-month Treasury yield spread, lagged two weeks
09. Baa/10-year Treasury yield spread, lagged two weeks
10. WTI weekly percentage change, lagged two weeks
11. WTI price level, lagged two weeks
12. Broad U.S. dollar weekly percentage change, lagged two weeks
13. Initial-claims weekly percentage change, lagged two weeks
14. Weekly Economic Index level, lagged two weeks
15. National Financial Conditions Index level, lagged two weeks
16. S&P 500 forward 12-month P/E level, lagged two weeks

C. **Threshold Features (16 conditions; two features each = 32)** – Each condition contributes a binary flag (1 when true, otherwise 0) and an excess term (the nonnegative distance beyond the cutoff, otherwise 0).
- For a below-cutoff condition: excess = max(cutoff - value, 0)
- For an above-cutoff condition: excess = max(value - cutoff, 0)

01. S&P 500 weekly return below -2%: flag and excess
02. S&P 500 weekly return above +2%: flag and excess
03. VIX weekly change below -5 points: flag and excess
04. VIX weekly change above +5 points: flag and excess
05. WTI weekly change below -5%: flag and excess
06. WTI weekly change above +5%: flag and excess
07. Dollar weekly change below -1%: flag and excess
08. Dollar weekly change above +1%: flag and excess
09. Initial-claims weekly change below -5%: flag and excess
10. Initial-claims weekly change above +5%: flag and excess
11. 10-year/3-month Treasury yield spread below zero: flag and excess
12. Weekly Economic Index below zero: flag and excess
13. Forward P/E below 16: flag and excess
14. Forward P/E above 20: flag and excess
15. VIX above 25: flag and excess
16. Baa/10-year Treasury yield spread above 250 basis points: flag and excess

No explicit timing-by-economic-regime interaction features were retained.

Training-Period Cross-Validation Results – Results below are means across the five chronological validation folds.
- These are validation metrics, not full-training in-sample performance.
- MAE and RMSE use decimal-return units: 0.0202 equals 2.02 percentage points of return error. R-squared and correlations are unitless.
- Values in this section are rounded to four decimal places.

Training-Period Cross-Validation Results (continued)

Training-Period Cross-Validation Results	
Model / benchmark	**Mean validation MAE**
Historical mean	0.0201
Zero return	0.0202
Gradient Boosting	0.0202
Elastic Net	0.0203
Ridge	0.0204
Random Forest	0.0208

Selected Gradient Boosting Model	
Mean Validation MAE	0.0202
Validation MAE Standard Deviation	0.0064
Mean Validation RMSE	0.0249
Mean Validation R-Squared	-0.0757
Notes No excessive training-versus-validation gap was flagged. Gradient Boosting had the lowest mean validation MAE among the four tuned models, but did not beat the historical-mean benchmark.	

Frozen-Model Test Results (184 Observations)	
MAE	0.0147500211
RMSE	0.0195008639
R-Squared	-0.0319046458
Directional Accuracy	54.35%
Pearson Correlation*	0.0621094
Historical-Mean Benchmark MAE	0.0145044314
Notes *Predicted versus actual next-week returns. The model's test MAE was below zero return and above the historical mean. Previously viewed test results are not fresh blinded validation. Notebook corrections did not use test performance to retune or reselect the model.	

Investment Simulation Results – Explanation of investment simulation for testing period.

- Decision dates: March 24, 2023-September 25, 2026; 184 opportunities.
- Final realized return and portfolio valuation: October 2, 2026.
- Rule: contribute $100 after a positive prediction; otherwise make no new contribution. Existing holdings remain invested. The reported simulation additionally forces the first $100 contribution despite a negative signal.

Investment Simulation Results		
Measure	**Model Strategy**	**Weekly $100 Benchmark**
Number of Contributions	132	184
Total contributions	$13,200	$18,400
Ending portfolio value	$18,333	$25,375
Gain above contributions	$5,133	$6,975
Money-weighted annualized return	18.86%	18.84%
Displayed annualized return	18.86%	18.84%
Notes Money-weighted annualized-return difference: approximately 0.0154 percentage points, or 0.02 points between separately rounded rates. Contributions were 28.26% lower than the weekly benchmark.		

Source & Scope

- Source: SP500_Capstone_Modeling V3.ipynb, completed V9 study.
- This summary does not change the model, data, notebook, or investment rules.
- The October 2 forecast for October 9 is not included as a performance result.